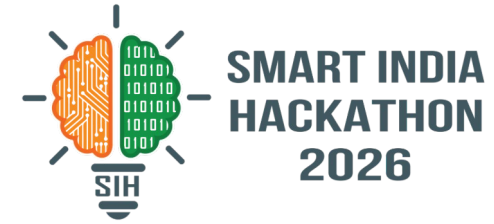
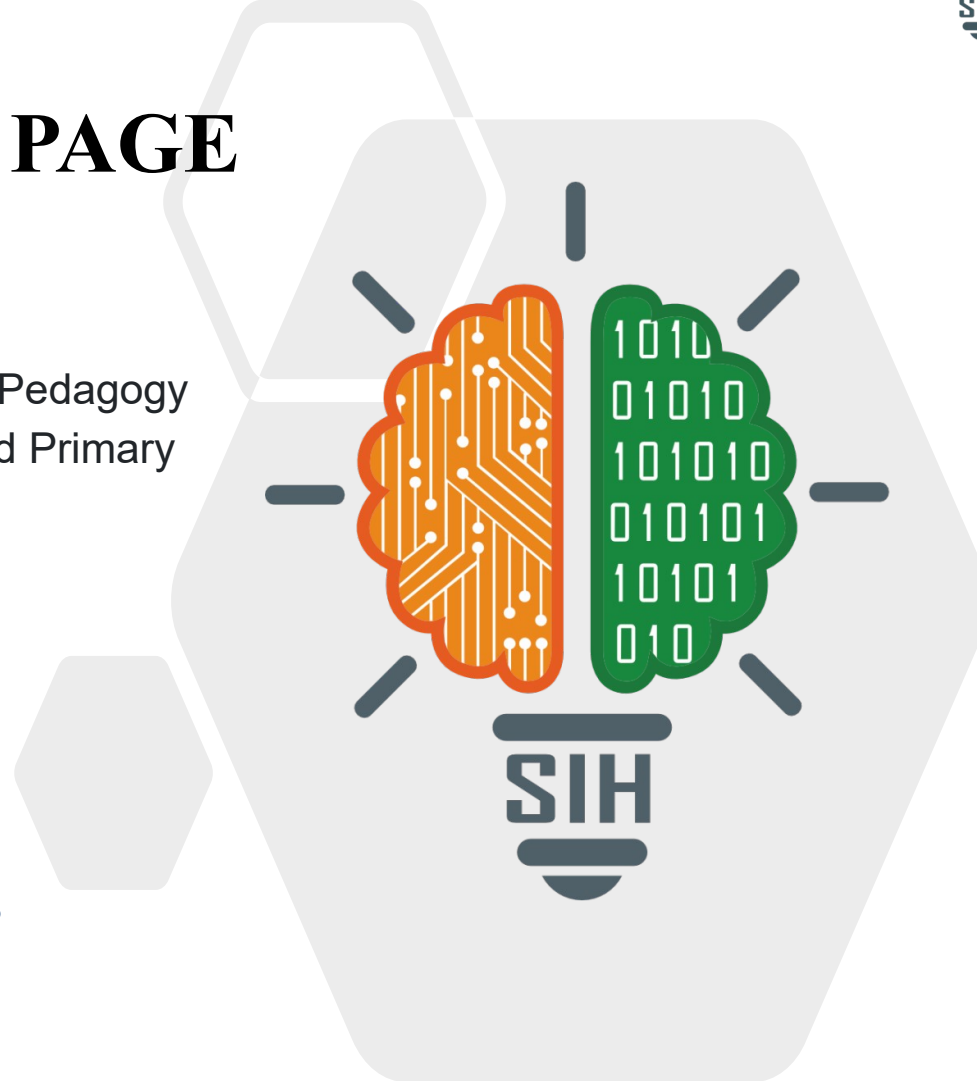


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26042 / 042
- **Problem Statement Title** – AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education
- **Theme** – Education / EdTech
- **PS Category** – Software
- **Team ID** – [ENTER YOUR OFFICIAL SIH TEAM ID]
- **Team Name** (Registered on portal) – Team Bytes



❖ ❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed Explanation of the Proposed Solution

- **Platform Overview:** AI-powered mobile/web platform translating primary educational content into tribal & regional mother tongues in real time.
- **Multimodal Core:** Combines speech-to-text (STT), text-to-speech (TTS), and neural machine translation (NMT) for two-way native interaction.
- **Offline Pedagogic Flow:** Generates bilingual teaching materials (synchronized text + audio) with offline-first caching for zero-connectivity classrooms.

• How It Addresses the Problem

- **Bridging Curriculum Gap:** Eliminates learning gaps between standardized state curricula and mother-tongue primary students, directly supporting NEP 2020.
- **Teacher Empowerment:** Provides educators with instant automated localization tools, removing dependency on scarce tribal-language human translators.

• Innovation and Uniqueness of the Solution

- **Modular Language Plugins:** Modular plug-and-play language architecture allows rapid addition of new tribal/regional dialects without modifying core engine.
- **Edge-Optimized Inference:** Quantized offline on-device AI inference engineered specifically for ultra-low-cost Android devices, unlike cloud-bound tools.

TECHNICAL APPROACH

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

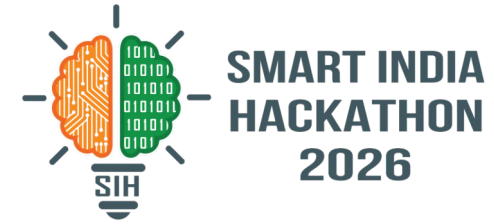
FEASIBILITY AND VIABILITY

- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

IMPACT AND BENEFITS

- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

RESEARCH AND REFERENCES



- Details / Links of the reference and research work